

Air Quality Analysis of Indian Cities (2015–2020)

An Exploratory Data Analysis

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing air quality data using Python (Pandas, Matplotlib, and Seaborn) to derive meaningful insights about pollution patterns across Indian cities.

The main objective is to demonstrate end-to-end exploratory data analysis (EDA) — from data cleaning to deriving actionable insights through visualisations and statistical summaries — using real government-sourced pollution data.

Objectives

- To identify meaningful patterns, trends, and insights from the air quality dataset using exploratory data analysis (EDA).
- To clean, transform, and preprocess the air quality data (handling missing values, converting date formats, and creating new columns like month and season) for analysis.
- To create meaningful visualisations that clearly show pollution trends across different cities, months, and seasons.
- To provide insights on which cities have the worst air quality and how pollution levels change over time.

Outcome

This project helps uncover key patterns and trends in air pollution across major Indian cities between 2015 and 2020. The analysis shows which cities have consistently poor air quality, how pollution levels change with seasons (like higher pollution in winter), and how different pollutants are related to each other. This gives a clearer understanding of India's air quality situation and can support better awareness and decision-making around pollution control.

2. Data Sources

- **Source:** Kaggle — "Air Quality Data in India", compiled from CPCB (Central Pollution Control Board), a Government of India monitoring body.
- **Timeline:** 2015 to 2020 (5 years of daily station-wise data).
- **Domain / Sub-domain:** Environmental Analytics — Air Quality Monitoring & Analysis.

3. Problem Statement

- To analyse air pollution patterns across 26 Indian cities to identify the most polluted regions and pollution trends.

- To study seasonal variation in pollution levels to understand when air quality is worst.
- To evaluate which pollutants are most strongly associated with overall AQI, to guide pollution-control priorities.
- To track how air quality has changed year-over-year from 2015 to 2020.

4. Attribute (Column / Feature) Details

Attribute Name	Data Type	Description
City	Categorical (Text)	Name of the Indian city where air quality was measured
Date	Date	The date on which the readings were recorded
PM2.5	Numeric (Float)	Fine particulate matter concentration
PM10	Numeric (Float)	Coarser particulate matter concentration
NO	Numeric (Float)	Nitric Oxide level, mainly from vehicle emissions
NO2	Numeric (Float)	Nitrogen Dioxide level, mainly from vehicles and industry
NOx	Numeric (Float)	Total nitrogen oxides in the air
NH3	Numeric (Float)	Ammonia level, mainly from agriculture and industry
CO	Numeric (Float)	Carbon Monoxide level, from burning fuel/vehicles
SO2	Numeric (Float)	Sulfur Dioxide level, mainly from burning coal/diesel
O3	Numeric (Float)	Ground-level Ozone concentration
Benzene	Numeric (Float)	Chemical pollutant from vehicle fuel and industry
Toluene	Numeric (Float)	Chemical pollutant, similar source as Benzene
AQI	Numeric (Float)	Air Quality Index - combined score representing overall pollution level
AQI_Bucket	Categorical (Text)	Category label based on AQI score (Good, Satisfactory, Moderate, Poor, Very Poor, Severe)
Year / Month / Season	Derived Columns	Extracted/engineered from Date during pre-processing, used for trend and seasonal analysis

5. Tools & Technologies

- **Python:** Google Colab (Jupyter Notebook) as the development environment.
- **Libraries:** Pandas and NumPy for data cleaning, transformation, and aggregation (groupby, pivot tables).
- **Visualisation:** Matplotlib and Seaborn for all visualisations (bar, line, box, pie, heatmap, scatter, subplots).
- **Export:** Exported the cleaned dataset as a CSV for future dashboard use.
- **Power BI:** To build interactive Dashboard

6. Data Pre-Processing (Python / Pandas)

Tasks Performed:

- Checked for duplicate rows (0 duplicates found) and missing values in each column, converted to percentages to judge severity.
- Converted the Date column from text to a proper datetime format.
- Dropped the Xylene column entirely, since 61% of its values were missing — too unreliable to impute.
- Filled missing values in other pollutant columns (PM2.5, PM10, NO2, etc.) using each city's own median, since pollution levels vary significantly by location, and the median avoids distortion from extreme spikes.
- Removed rows with missing AQI/AQI_Bucket, since AQI is a calculated composite index and cannot be reliably estimated from already-imputed values.
- Created derived columns — Year, Month, Month_Name, and Season (Winter/Summer/Monsoon/Post-Monsoon) — to enable time-based and seasonal analysis.

7. Skewness and Outlier Analysis

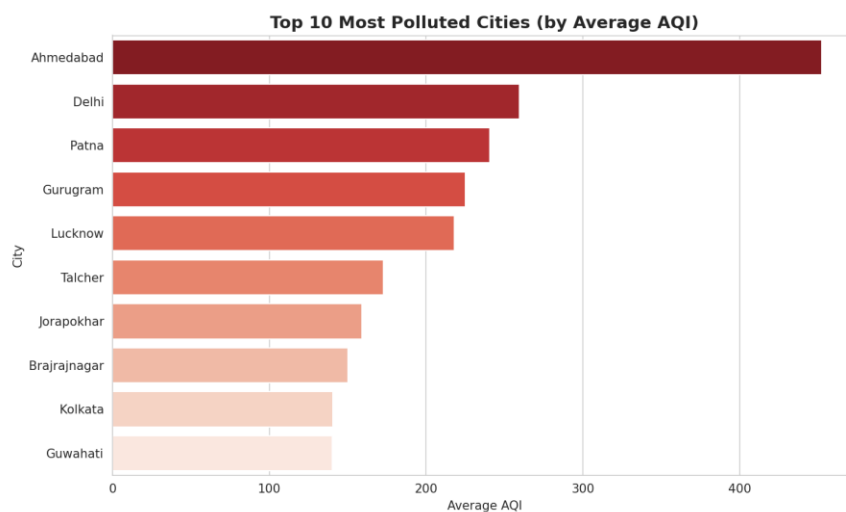
Before EDA, we checked numerical columns for skewness and outliers to understand the underlying distribution of the pollution data.

- All pollutant columns and AQI showed strong positive (right) skewness, ranging from 1.35 (O3) up to 23.25 (Benzene) — meaning most days have moderate pollution, with occasional extreme spikes pulling the distribution's tail to the right.
- Using the IQR method, 3% to 12% of values per column were flagged as statistical outliers.
- Minimum and maximum values were checked to confirm these were genuine pollution events (e.g. PM2.5 up to 914.94, PM10 up to 917.08) rather than data errors, matching real severe pollution events recorded in Indian cities.
- Decision: outliers were kept, not removed, since they represent real, meaningful pollution spikes (festivals, stubble burning, winter smog) central to this analysis, rather than data entry errors.

8. Analysis and Visualisations (Python)

Type of Analysis: This project focuses on Descriptive Analysis (summarising pollution levels and trends) and Diagnostic Analysis (identifying why certain cities/seasons show higher pollution).

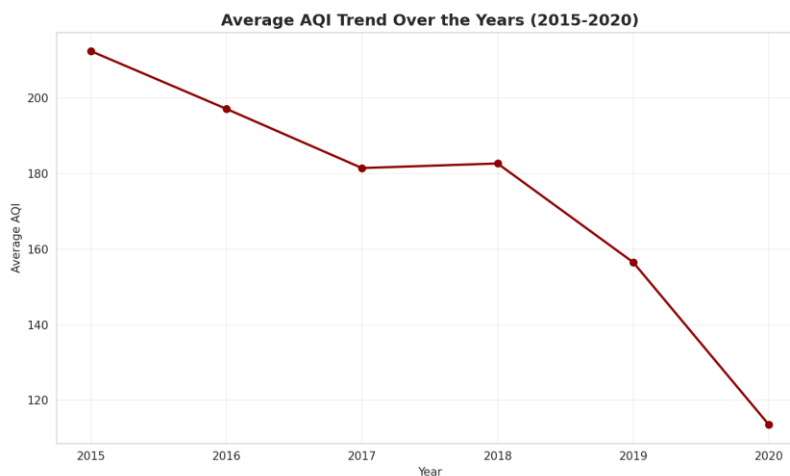
Analysis 1: Most Polluted Cities in India (Bivariate)



Top 10 most polluted Indian cities by average AQI (2015-2020)

- Ahmedabad (avg. AQI 452) and Delhi (avg. AQI 260) are by far the most polluted cities, followed by Patna, Gurugram, and Lucknow.

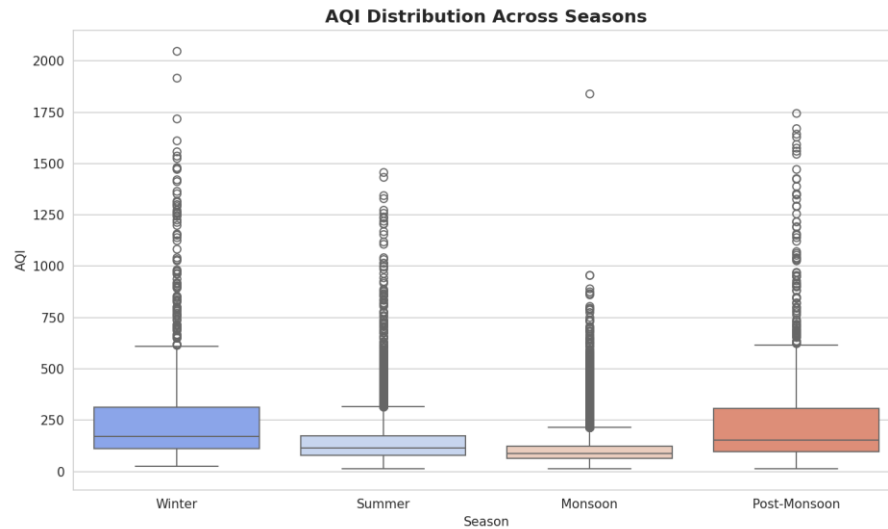
Analysis 2: AQI Trend Over the Years (Bivariate)



Average AQI trend across India, year by year

- Average AQI declined steadily from 212 (2015) to 114 (2020), suggesting an overall improvement in air quality over the 5 years.

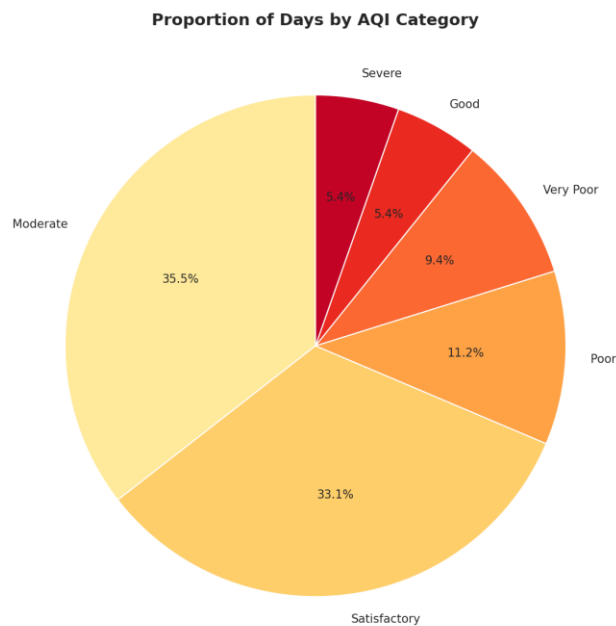
Analysis 3: AQI Distribution Across Seasons (Bivariate)



Spread of AQI values across the four seasons

- Winter shows the highest average AQI (221), closely followed by Post-Monsoon (216), while Monsoon is the cleanest season (116) - likely due to rain washing pollutants from the air.

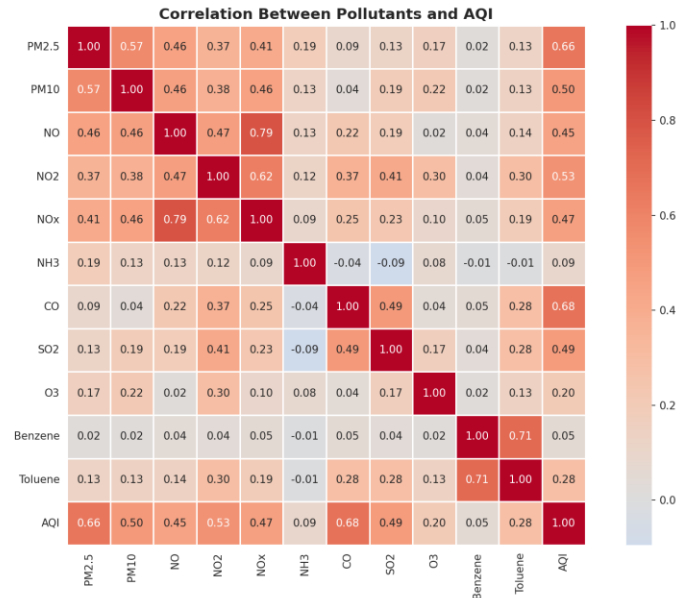
Analysis 4: AQI Category Distribution (Univariate)



Proportion of days falling into each AQI category

- Only 5.4% of days were categorised as "Good", while over 26% of days fell into Poor, Very Poor, or Severe categories, showing unhealthy air is a frequent occurrence, not a rare one.

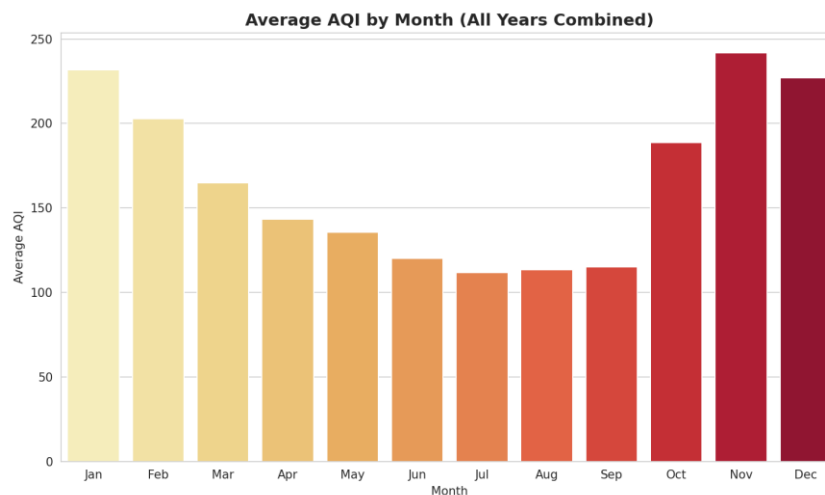
Analysis 5: Pollutant Correlation with AQI (Multivariate)



Correlation heatmap between all pollutants and AQI

- CO (correlation 0.68) and PM2.5 (correlation 0.66) show the strongest relationship with AQI, confirming these as the main drivers of poor air quality days.

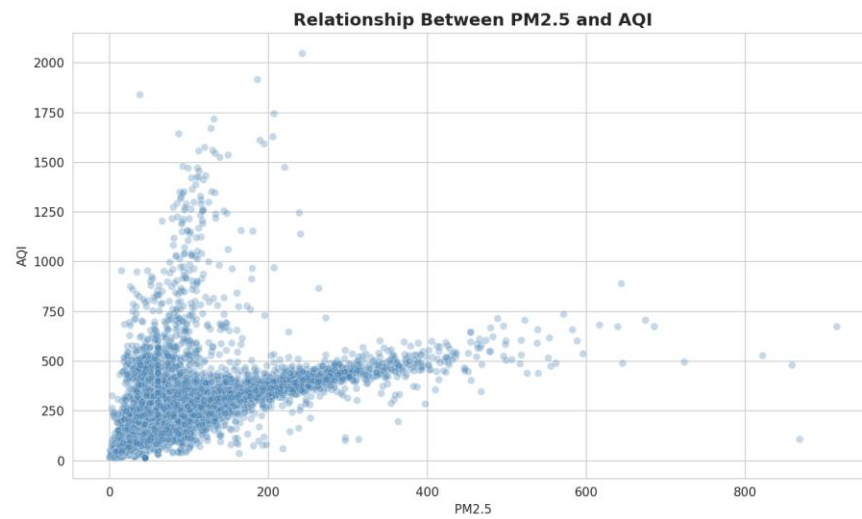
Analysis 6: AQI by Month (Bivariate)



Average AQI for each calendar month, combined across all 5 years

- Winter months (November-January) consistently show higher average AQI, reinforcing the seasonal pattern seen in Analysis 3.

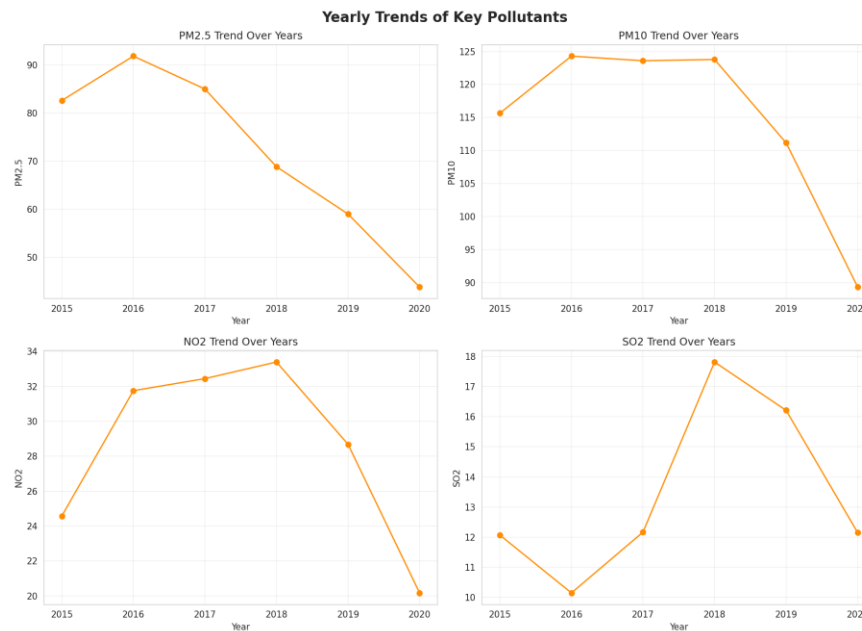
Analysis 7: PM2.5 and AQI Relationship (Bivariate)



Relationship between PM2.5 concentration and overall AQI

- A clear positive relationship is visible - as PM2.5 increases, AQI increases correspondingly, visually confirming PM2.5 as a leading driver of poor air quality.

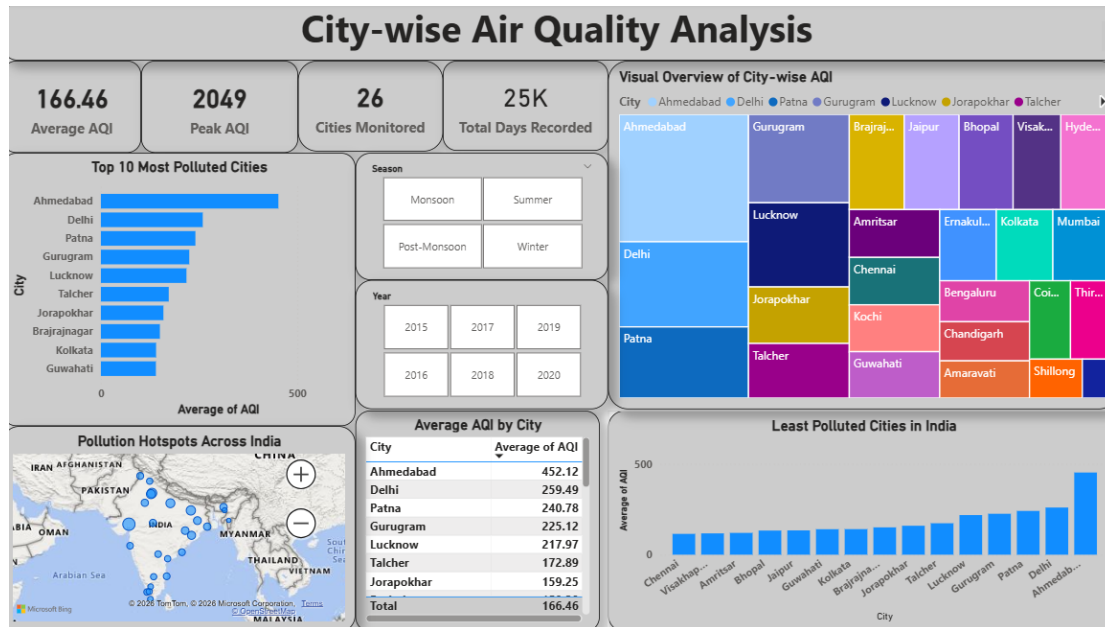
Analysis 8: Yearly Trends of Key Pollutants (Multivariate)



Year-wise trend of PM2.5, PM10, NO2, and SO2

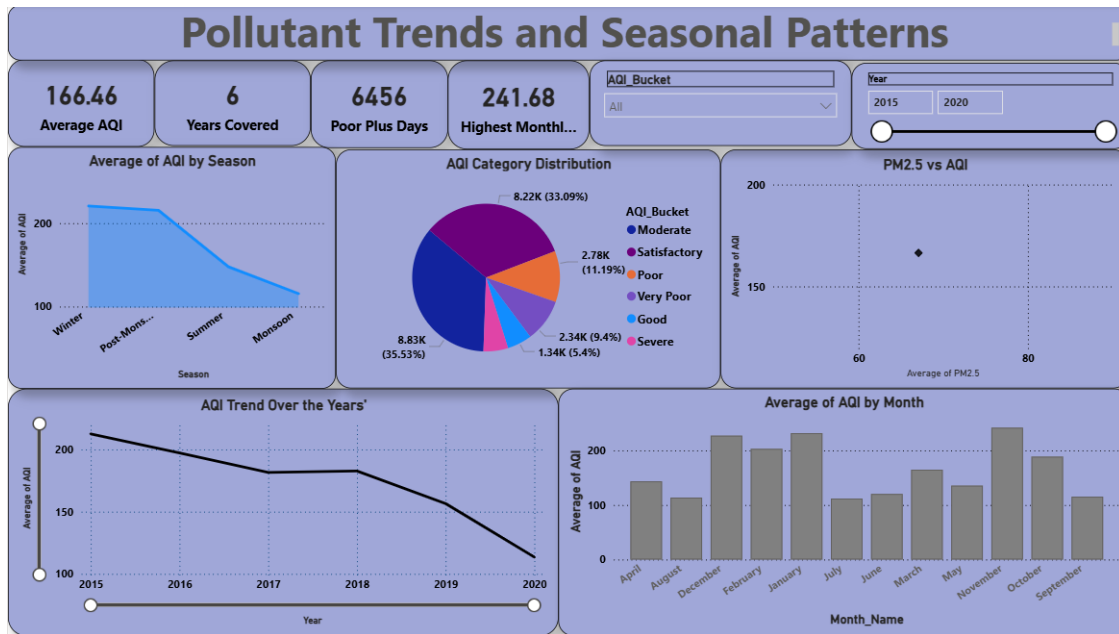
- Individual pollutant trends can be compared side-by-side, showing that most key pollutants follow a declining pattern over the years, consistent with the overall AQI improvement.

9. Analysis and Visualisations (Power BI)



About the Dashboard:

- This dashboard was built in Power BI using the same cleaned dataset (24,850 records, 26 cities), and focuses on comparing pollution levels across cities, rather than across time.
- Four KPI cards give a quick summary: Average AQI (166.46) across the whole dataset, the single worst reading recorded, 26 Cities Monitored, and ~25,000 Total Days Recorded. Peak AQI (2049)
- The bar chart and table both confirm the same finding from a different angle: Ahmedabad (452.12 avg AQI) and Delhi (259.49 avg AQI) are by far the most polluted cities, followed by Patna, Gurugram, and Lucknow.
- The Treemap gives a quick visual scan of all 26 cities at once, sized/colored by their median AQI — making it easy to spot Ahmedabad, Delhi, and Patna standing out immediately.
- The Map visual shows these pollution hotspots are geographically spread across India, not localised to one region — from Delhi/NCR in the north, to Kolkata in the east, to cities further south.
- Season and Year slicers let a viewer interactively filter all these visuals — for example, checking whether Ahmedabad stays the worst city even in Monsoon season specifically.



About the Dashboard:

- This dashboard shifts focus from *where* pollution is worst to *when* and *why* — looking at time-based and category-based patterns using the same cleaned dataset.
- The 4 KPI cards show: Average AQI (166.68), data spanning 6 Years (2015-2020), 6,456 "Poor Plus" Days (days categorised as Poor, Very Poor, or Severe combined), and a Highest Monthly Average AQI of 241.68 (November, the worst month on average).
- The Season chart confirms Winter and Post-Monsoon have the highest average AQI, while Monsoon is clearly the cleanest season — matching the seasonal pattern already seen in the Python analysis.
- The Month-wise chart adds finer detail: November, December, and January stand out as the worst months, while June-September (monsoon months) are consistently lower.
- The AQI Category pie chart shows the real scale of the problem: only 5.4% of days were "Good", while Poor + Very Poor + Severe together made up about 26% of all recorded days (6,456 out of 24,850) — meaning roughly 1 in 4 days had genuinely unhealthy air.
- The Year-over-year line chart shows Average AQI declining from ~212 in 2015 to ~114 in 2020, suggesting a real improvement in air quality over the 5 years.

10. Insights & Conclusions

Key Findings (Descriptive & Diagnostic Analysis)

- Ahmedabad and Delhi are India's most polluted cities in this dataset, with average AQI of 452 and 259 respectively, far above the next closest city, Patna (241).
- Winter is consistently the worst season for air quality, while Monsoon is the cleanest a pattern that shows up both in the season chart and the month-wise breakdown (Nov-Jan are the worst months).
- Air quality has improved over time — average AQI dropped from around 212 in 2015 to 114 in 2020, a meaningful downward trend across the 5-year period.
- Unhealthy air is common, not rare — about 1 in 4 recorded days (6,456 out of 24,850) fall into Poor, Very Poor, or Severe categories.
- Only 5.4% of days were categorised as "Good" — showing that genuinely clean air is the exception, not the norm, across this dataset.
- PM2.5 and CO are the pollutants most strongly linked to overall AQI (correlation of 0.66 and 0.68 respectively, from the correlation heatmap), making them useful early indicators of a bad air quality day.
- The single worst AQI recorded was 2049 — notably above the standard 0-500 AQI scale, flagged as an extreme outlier event rather than a typical day.
- Pollution hotspots are spread across India geographically, not limited to one region — the map shows affected cities from the north (Delhi/Gurugram) to the east (Kolkata) to other states.
- 26 cities and nearly 25,000 days of data gave enough coverage to confidently compare cities and track patterns over multiple years, not just a single snapshot.

11. Conclusion

If a business or government body were to use this analysis, here's what it tells them: certain cities — especially Ahmedabad, Delhi, and Patna — need priority attention, since their pollution levels are dramatically higher than the rest of the dataset, not just slightly worse. Winter is the critical intervention window — any pollution-control measures (traffic restrictions, construction bans, stubble-burning enforcement) would have the most impact if timed around November to January, rather than spread evenly across the year. On a positive note, the declining AQI trend from 2015 to 2020 suggests existing efforts may already be working, and this can be used as a benchmark to measure whether future interventions keep improving air quality or cause it to plateau/worsen again. Since PM2.5 and CO show the strongest link to overall AQI, monitoring and controlling these two pollutants specifically would likely give the most "bang for the buck" in terms of improving the overall AQI score. Finally, since almost a quarter of all days are unhealthy, this isn't a rare-emergency problem — it's a frequent, ongoing issue that needs sustained, year-round monitoring rather than reactive one-off responses.